1. Let $I(x)=\int \frac{x^2\left(x \sec^2 x+\tan x\right)}{(x \tan x+1)^2} d x$ If $I(0)=0$, then $I\left(\frac{\pi}{4}\right)$ is equal to

[2023 (06 Apr Shift 1)]

(1) $\log_e \frac{(\pi+4)^2}{16}+\frac{\pi^2}{4(\pi+4)}$

(2) $\log_e \frac{(\pi+4)^2}{16}-\frac{\pi^2}{4(\pi+4)}$

(3) $\log_e \frac{(\pi+4)^2}{32}-\frac{\pi^2}{4(\pi+4)}$

(4) $\log_e \frac{(\pi+4)^2}{32}+\frac{\pi^2}{4(\pi+4)}$

2. Let $I(x)=\int \frac{x+1}{x\left(1+x e^x\right)^2} d x,\ x>0$. If $\lim_{x \rightarrow \infty} I(x)=0$ then $I(1)$ is equal to

[2023 (08 Apr Shift 1)]

(1) $\frac{e+2}{e+1}-\log_e(e+1)$

(2) $\frac{e+1}{e+2}+\log_e(e+1)$

(3) $\frac{e+1}{e+2}-\log_e(e+1)$

(4) $\frac{e+2}{e+1}+\log_e(e+1)$

3. The integral $\int\left(\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x\right) \log_2 x\, d x$ is equal to

[2023 (08 Apr Shift 2)]

(1) $\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x+C$

(2) $\left(\frac{x}{2}\right)^x-\left(\frac{2}{x}\right)^x+C$

(3) $\left(\frac{x}{2}\right)^x \log_2\left(\frac{x}{2}\right)+C$

(4) $\left(\frac{x}{2}\right)^x \log_2\left(\frac{2}{x}\right)+C$

4. If $I(x)=\int e^{\sin^2 x}(\cos x \ \sin 2 x-\sin x) d x$ and $I(0)=1$, then $I\left(\frac{\pi}{3}\right)$ is equal to

[2023 (10 Apr Shift 1)]

(1) $\frac{1}{2} e^{\frac{3}{4}}$

(2) $\frac{1}{2} e^{\frac{3}{4}}$

(3) $-e^{\frac{3}{4}}$

(4) $e^{\frac{3}{4}}$

5. For $\alpha, \beta, \gamma, \delta \in \mathbb{N}$, if $\int\left(\left(\frac{x}{e}\right)^{2 x}+\left(\frac{e}{x}\right)^{2 x}\right) \log_e x\, d x=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$, where $e=\sum_{n=0}^{\infty} \frac{1}{n!}$ and C is constant of integration, then $\alpha+2 \beta+3 \gamma-4 \delta$ is equal to

[2023 (10 Apr Shift 2)]

(1) 1

(2) 4

(3) -4

(4) -8

6. Let $I(x)=\int \sqrt{\frac{x+7}{x}}\, d x$ and $I(9)=12+7 \log_e 7$. If $I(1)=\alpha+7 \log_e(1+2 \sqrt{2})$, then α^4 is equal to _____.

[2023 (12 Apr Shift 1)]

7. $\int_0^{\infty} \frac{6}{e^{3 x}+6 e^{2 x}+11 e^x+6} d x=$

[2023 (13 Apr Shift 1)]

(1) $\log_e\left(\frac{32}{27}\right)$

(2) $\log_e\left(\frac{512}{81}\right)$

(3) $\log_e\left(\frac{256}{81}\right)$

(4) $\log_e\left(\frac{302}{27}\right)$

8. Let $f(x)=\int \frac{d x}{\left(3+4 x^2\right) \sqrt{4-3 x^2}},\ |x|<\frac{2}{\sqrt{3}}$. If $f(0)=0$ and $f(1)=\frac{1}{\alpha \beta} \tan^{-1}\left(\frac{\alpha}{\beta}\right),\ \alpha,\ \beta>0$, then $\alpha^2+\beta^2$ is equal to _______.

[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (3) **2.** (1) **3.** (1) **4.** (2) **5.** (2) **6.** (64) **7.** (1) **8.** (28)

1. (3)

Given that $I(x) = \int x^2 \left(\frac{x \sec^2 x + \tan x}{(x \tan x + 1)^2}\right) dx$

Apply Integration by parts.

$\int f(x) g(x) dx = f(x) \int g(x) dx - \int \left(f'(x) \int g(x) dx\right) dx$

$= x^2 \int \left(\frac{x \sec^2 x + \tan x}{(x \tan x + 1)^2}\right) dx - \int \left(\frac{\mathrm{d} x^2}{\mathrm{d} x} \int \left(\frac{x \sec^2 x + \tan x}{(x \tan x + 1)^2}\right) dx\right) dx$

Let $x \tan x + 1 = p$

$\Rightarrow (x \sec^2 x + \tan x) dp = dx$

$= x^2 \int \frac{dp}{p^2} - \int \left(2x \int \frac{dp}{p^2}\right) dx$

$= \frac{-x^2}{(x \tan x + 1)} + \int \frac{2x}{x \tan x + 1} dx$

Let $I_1 = 2 \int \frac{x}{x \tan x + 1} dx$

$= 2 \int \frac{x \cos x}{x \sin x + \cos x} dx$

Let $x \sin x + \cos x = t$

$\Rightarrow (x \cos x + \sin x - \sin x) dx = dt$

$\Rightarrow (x \cos x) dx = dt$

$= 2 \int \frac{dt}{t} = 2 \log t + c$

$= 2 \log |x \sin x + \cos x| + c$

$\therefore \int \frac{x^2 (x \sec^2 x + \tan x)}{(x \tan x + 1)^2} dx$

$= \frac{-x^2}{x \tan x + 1} + 2 \log \left| x \sin x + \cos x \right| + c$

But $I(0) = 0$

$\Rightarrow c = 0$

Also, $I\left(\frac{\pi}{4}\right) = -\frac{\left(\frac{\pi}{4}\right)^2}{\frac{\pi}{4} \times 1 + 1} + 2 \log \left| \frac{1}{\sqrt{2}} \left(\frac{\pi}{4}\right) + \frac{1}{\sqrt{2}} \right|$

$= \log_e \frac{(\pi + 4)^2}{32} - \frac{\pi^2}{4(\pi + 4)}$

Hence, this is the correct option.

2. (1)

Given,

$I(x) = \int \frac{x+1}{x(1 + x e^x)^2} dx$

Now let $1 + x e^x = t$

$\Rightarrow e^x (x + 1) dx = dt$

So, $I(x) = \int \frac{1}{(t-1) t^2} dt$

$\Rightarrow I(x) = \int \frac{(1 - t^2) + t^2}{(t-1) t^2} dt$

$\Rightarrow I(x) = \int \frac{-(t+1)}{t^2} + \frac{1}{t-1} dt$

$\Rightarrow I(x) = \int -\frac{1}{t} - \frac{1}{t^2} + \frac{1}{t-1} dt$

$\Rightarrow I(x) = -\ln t + \frac{1}{t} + \ln(t - 1) + C$

$\Rightarrow I(x) = \ln\left(\frac{x e^x}{x e^x + 1}\right) + \frac{1}{x e^x + 1} + C$

Also given, $\lim_{x \to \infty} I(x) = 0$

$\Rightarrow \lim_{x \to \infty} I(x) = \lim_{x \to \infty} \left[\ln\left(1 - \frac{1}{x e^x + 1}\right) + \frac{1}{x e^x + 1} + C\right]$

$\Rightarrow \lim_{x \to \infty} I(x) = [\ln(1 - 0) + 0 + C]$

$\Rightarrow C = 0$

Now finding,

$I(1) = \ln\left(\frac{e}{e+1}\right) + \frac{1}{e+1} = 1 + \frac{1}{e+1} - \ln(e + 1)$

$\Rightarrow I(1) = \frac{e+2}{e+1} - \ln(e + 1)$

3. (1)

Given,

$I=\int\left(\left(\frac{x}{2}\right)^{x}+\left(\frac{2}{x}\right)^{x}\right)\log_2 x\,dx$

Now, Let $\left(\frac{x}{2}\right)^{x}=t$

$\Rightarrow\ x\log_2\left(\frac{x}{2}\right)=\log_2 t$

$\Rightarrow\ x\log_e\left(\frac{x}{2}\right)\cdot\log_2 e=\log_e t\cdot\log_2 e$

On differentiating both sides we get :

$\log_e\left(\frac{x}{2}\right)+x\cdot\frac{2}{x}\cdot\frac{1}{2}=\frac{1}{t}\frac{dt}{dx}$

$\Rightarrow\log_e\left(\frac{x}{2}\right)+1=\frac{1}{t}\frac{dt}{dx}$

Then solution is not possible as there is no proper substitution.

Note: This question was bonus in Jee Mains 2023 April session.

4. (2)

Given,

$I(x)=\int e^{\sin^2 x}(\cos x\ \sin 2x-\sin x)dx$

$\Rightarrow I(x)=\int e^{\sin^2 x}\left(\cos x-\frac{1}{2\cos x}\right)\sin 2x\,dx$

Now let, $\sin^2 x=t\Rightarrow\sin 2x\,dx=dt$

$I(x)=\int e^{t}\left(\sqrt{1-t}-\frac{1}{2\sqrt{1-t}}\right)dt$

Now comparing with,

$I(x)=\int e^{t}(f(t)+f'(t))dt=e^{t}f(t)+c$

Here, $f(t)=\sqrt{1-t}$ & $f'(t)=-\frac{1}{2\sqrt{1-t}}$

$\Rightarrow I(x)=e^{t}\sqrt{1-t}+c=e^{\sin^2 x}\cdot\cos x+c$

Now using, $I(0)=1\Rightarrow c=0$

Hence, the value of $I\left(\frac{\pi}{3}\right)=\frac{1}{2}e^{\frac{3}{4}}$

5. (2)

Given,

$\int\left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\ln x\,dx=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$

Now let $I=\int\left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\ln x\,dx$

Now let $\left(\frac{x}{e}\right)^{2x}=t$

$\Rightarrow 2x(\ln x-1)=\ln t$

$\Rightarrow\ln x\,dx=\frac{1}{2t}dt$

So, $I=\frac{1}{2}\int\left(t+\frac{1}{t}\right)\frac{dt}{t}$

$\Rightarrow I=\frac{1}{2}\int\left(1+\frac{1}{t^2}\right)dt$

$\Rightarrow I=\frac{1}{2}\left(t-\frac{1}{t}\right)+c$

$\Rightarrow I=\frac{1}{2}\left(\left(\frac{x}{e}\right)^{2x}-\left(\frac{e}{x}\right)^{2x}\right)+C$

Now on comparing with $I=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$, we get $\alpha=2,\ \beta=2,\gamma=2,\delta=2$

Hence, $\alpha+2\beta+3\gamma-4\delta=2+4+6-8=4$

Hence this is the required option.

6. **(64)**

Given,

$I(x)=\int\sqrt{\frac{x+7}{x}}\,dx$

Now let, $\frac{x+7}{x}=t^2\Rightarrow-\frac{7}{x^2}dx=2t\,dt$

$\Rightarrow dx=\frac{-14t}{\left(t^2-1\right)^2}dt$

So, $I(x)=-14\int\frac{t^2}{\left(t^2-1\right)^2}dt$

$\Rightarrow I(x)=-14\int\frac{dt}{\left(t^2+\frac{1}{t^2}-2\right)}$

$\Rightarrow I(x)=\frac{-14}{2}\int\left[\frac{\left(1-\frac{1}{t^2}\right)}{\left(t+\frac{1}{t}\right)^2-4}+\frac{\left(1+\frac{1}{t^2}\right)}{\left(t-\frac{1}{t}\right)^2}\right]dt$

$\Rightarrow I(x)=-7\left(\frac{1}{4}\ln\left|\frac{t+\frac{1}{t}-2}{t+\frac{1}{t}+2}\right|-\frac{1}{t-\frac{1}{t}}\right)+c$

Now when $x=9,\ t=\frac{4}{3}$

$\Rightarrow I(9)=\ 12+7\times\ln 7=\frac{-7}{4}\ln\left(\frac{1}{7}\right)^2+7\times\frac{12}{7}+c$

$\Rightarrow\ c=\frac{7}{2}\ln 7$

Now when $x=1,\ t=2\sqrt{2}$

$\Rightarrow\ I(1)=+\frac{7}{4}\ln\left(\frac{2\sqrt{2}+1}{2\sqrt{2}-1}\right)^2+7\times\frac{2\sqrt{2}}{7}+\frac{7}{2}\ln 7$

$\Rightarrow I(1)=\frac{7}{2}\ln\left(\frac{\left(2\sqrt{2}+1\right)^2}{7}\right)+2\sqrt{2}+\frac{7}{2}\ln 7$

$\Rightarrow I(1)=7\ \ln\left(2\sqrt{2}+1\right)-\frac{7}{2}\ln 7+2\sqrt{2}+\frac{7}{2}\ln 7$

$\Rightarrow\ \alpha=2\sqrt{2}\Rightarrow\alpha^4=64$

7. **(1)**

Let

$I=\int_0^\infty\left(\frac{6}{e^{3x}+6e^{2x}+11e^x+6}\right)dx$

$\Rightarrow I=\int_0^\infty\frac{6}{\left(e^x+1\right)\left(e^x+2\right)\left(e^x+3\right)}dx$

Using partial fraction, we can write

$\Rightarrow I=6\int_0^\infty\left[\frac{1}{2\left(e^x+1\right)}-\frac{1}{\left(e^x+2\right)}+\frac{1}{2\left(e^x+3\right)}\right]dx$

$\Rightarrow I=6\int_0^\infty\left[\frac{1}{2\left(e^{-x}+1\right)}-\frac{1}{\left(1+2e^{-x}\right)}+\frac{1}{2\left(1+3e^{-x}\right)}\right]e^{-x}dx$

$\Rightarrow I=6\left[-\frac{1}{2}\log_e\left(e^{-x}+1\right)+\frac{1}{2}\log_e\left(2e^{-x}+1\right)-\frac{1}{6}\log_e\left(3e^{-x}+1\right)\right]_0^\infty$

$\Rightarrow I=6\left[0-\left\{-\frac{1}{2}\log_e 2+\frac{1}{2}\log_e 3-\frac{1}{6}\log_e 4\right\}\right]$

$\Rightarrow I=3\log_e 2-3\log_e 3+\log_e 4$

$\Rightarrow I=5\log_e 2-3\log_e 3$

$\Rightarrow I=\log_e\left(\frac{32}{27}\right)$

8. (28)

Given,

$f(x)=\int \frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$

Put $x=\frac{1}{t},\ dx=-\frac{1}{t^2}dt$

So, $f(x)=\int \frac{-dt}{t^2\left(3+\frac{4}{t^2}\right)\sqrt{4-\frac{3}{t^2}}}$

$\Rightarrow f(x)=\int \frac{-t\,dt}{\left(3t^2+4\right)\sqrt{4t^2-3}}$

Now let, $4t^2-3=\lambda^2 \Rightarrow 8t\,dt=2\lambda\,d\lambda$

$\Rightarrow f(x)=-\int \frac{\lambda\,d\lambda}{4\cdot\left(3\left(\frac{\lambda^2+3}{4}\right)+4\right)\cdot\lambda}$

$\Rightarrow f(x)=-\int \frac{d\lambda}{3\lambda^2+9+16}$

$\Rightarrow f(x)=-\int \frac{d\lambda}{3\lambda^2+25}$

$\Rightarrow f(x)=-\frac{1}{3}\int \frac{d\lambda}{\lambda^2+\frac{25}{3}}$

$\Rightarrow f(x)=-\frac{1}{3}\times\frac{\sqrt{3}}{5}\tan^{-1}\left(\frac{\sqrt{3}\lambda}{5}\right)+c$

$\Rightarrow f(x)=-\frac{\sqrt{3}}{15}\tan^{-1}\left(\frac{\sqrt{3}\sqrt{\left(4-3x^2\right)}}{5x}\right)+c$

Now using, $f(0)=0 \Rightarrow c=+\frac{\sqrt{3}\pi}{30}$

Hence, $f(1)=\frac{-\sqrt{3}}{15}\tan^{-1}\left(\frac{\sqrt{3}}{5}\right)+\frac{\sqrt{3}}{15}\times\frac{\pi}{2}$

$\Rightarrow f(1)=\frac{-\sqrt{3}}{15}\left(\tan^{-1}\left(\frac{\sqrt{3}}{5}\right)-\frac{\pi}{2}\right)$

$\Rightarrow f(1)=\frac{\sqrt{3}}{15}\left(\frac{\pi}{2}-\tan^{-1}\left(\frac{\sqrt{3}}{5}\right)\right)$

$\Rightarrow f(1)=\frac{\sqrt{3}}{15}\cot^{-1}\left(\frac{\sqrt{3}}{5}\right)$

$\Rightarrow f(1)=\frac{\sqrt{3}}{15}\tan^{-1}\left(\frac{5}{\sqrt{3}}\right)$

$\Rightarrow f(1)=\frac{1}{5\sqrt{3}}\tan^{-1}\left(\frac{5}{\sqrt{3}}\right)$

Now comparing with given value of $f(1)$ we get, $\alpha=5$ & $\beta=\sqrt{3}$

$\Rightarrow \alpha^2+\beta^2=28$